**8.c)** What are genetic markers and what is their usefulness? Why are blood groups considered as good genetic markers? Illustrate with examples. (15 marks)

Genetic markers refers to the particular sequence of genes or their expression whose presence or absence helps in differentiating different set of human populations.

Therefore they act as 'markers' (identifying features) of given population and also as 'marker' for human variations.

**Significance of genetic markers**

1) Genetic markers help in evolutionary studies. For example, Out of Africa theory is strongly supported by modern genetics as African population has most diverse set of genetic markers, capable of being parent population

2) Genetic markers - useful in demographic

studies - in determining the reproductive fitness of different populations

3) Genetic markers - crucial for medicine or medical anthropology - they help determine or diagnose various diseases

[Eg] Transferrin ↓ ⇒ Anaemic conditions

4) Nutritional anthropology - certain genetic markers determine the individual's capacity to consume & process certain food

[Eg] Lactose intolerance

**Blood groups considered as good marker**

Blood groups like ABO blood group system are considered 'good' because of following features

1) They are of distinct categories -

[Eg] A blood group → +ve or -ve
B blood group → +ve or -ve, etc

2) They are present in all human population
3) They don't change with time ~~or~~ or space i.e. they remain constant in human wherever they are
4) They are easy to determine - don't need complex or sophisticated system
5) They are affordable & cheaper ~~to us~~ - can be used widespread ~~at~~ by all countries.

Importance

1) They are used to study classification of human population both inter-national manner or intra national manner
2) They are extensively used for diagnosis of various diseases

eg erythroblastosis foetalis, etc

Therefore, genetic markers have become integral to both individual level human genetics & population genetics.